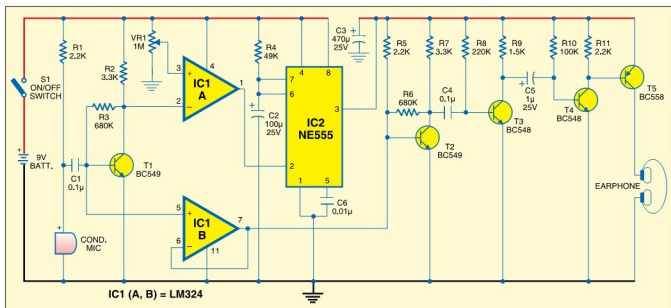




at its collector is fed to the inverting terminal (pin 2) of op-amp IC1(A), which is used as a comparator. The reference voltage (V_{ref}) at the non-inverting terminal (pin 3) of IC1(A) is set using preset VR1. The preset is also used to control the sensitivity of the sound signals received by the circuit. The output from pin 1 of IC1(A) is fed to the trigger input (pin 2) of timer NE555, which is configured in monostable mode.

When sufficient sound signal strength is detected at the base of transistor T1, the pulsating voltage at its collector exceeds the reference voltage at pin 3. As a result, output pin 1 of IC1(A) goes low. The low output from

IC1(A) triggers the NE555 timer and its output goes high for a preset duration. R4 and C2 are the timing components for setting the time duration. The high output of the timer is directly used as the power source for the sound amplifier section.

The sound amplifier section is built around transistors T2 through T5. The last amplifier stage T5 (pnp transistor BC558) drives the earphone. The sound signal received from the mic is fed to the non-inverting pin of the second op-amp of IC1(B) which is wired in unity follower mode. The unity follower mode resolves the problem of impedance mismatch which would have occurred if the output of the mic

is fed directly to amplifier stage. The output from pin 7 of IC1(B) is fed to the base of transistor T2. The weak signal received at transistor stage T2 is further amplified by transistors T3, T4 and T5. An earphone to listen to the sound is connected between the collector of T5 and ground. It is recommended to use a mono earphone with volume control attached.

With 9V DC supply, when sound is detected through the mic, the amplifier section is automatically triggered and the current consumption of the circuit is about 96 mA. When the amplifier circuit is 'off', the circuit draws a current of about 6 mA only, thus saving considerable amount of battery power. ●

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